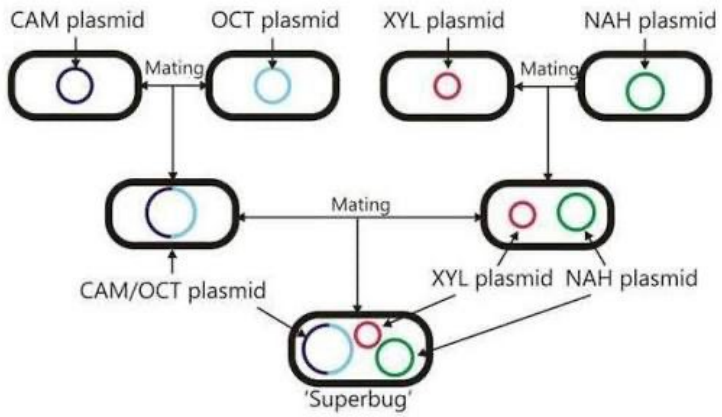
## Ananda Chakrabarty

For a subject rooted in clear-cut facts, science is accompanied surprisingly often by an ethical grey area. Ananda Chakrabarty's invention, while inarguably beneficial to the environment, left much of the public questioning and pushing back on the boundaries of this area.

Chakrabarty earned his PhD from the University of Calcutta in India, during which he published research on biochemistry and microbial genetics. Dr. Irwin Gunsalus was impressed by his work and invited him to study plasmid molecular biology in his lab at UIUC. This was where he developed his interest in Pseudomonas, the bacteria he spent the rest of his life experimenting on.

Chakrabarty then joined the General Electric Company in New York, where his project focused on using bacteria to convert cow manure to cattle feed. However, his interests didn't align exactly with this work so he took on a side project to address one of the biggest environmental concerns at the time: oil spills. Certain strains of Pseudomonas could naturally break down parts of crude oil, but each strain could only digest specific hydrocarbons. He discovered that by inserting multiple plasmids, each with genes for enzymes that break down different hydrocarbons, they could create a modified version of Pseudomonas that degrades crude oil.




This discovery carried significant promise in recovering the environment. However, the General Electric Company wanted to commercialize Chakrabarty's invention so they pressured him to file a patent on his bacteria. The main issue was that a patent had never been granted on a living organism before.

After almost a decade of legal battles, The Supreme Court decided 5-4 that genetically engineered organisms could be patented because they were human made products. This decision was revolutionary in the genetic engineering field and it's why Chakrabarty is often credited for the rise of the U.S biotech industry.

## Alice Ball

Imagine receiving news that a miraculous cure has been found for your death sentence. That was the case for millions suffering from Leprosy in the 20th century. The reason for this miracle? Alice Augusta Ball.

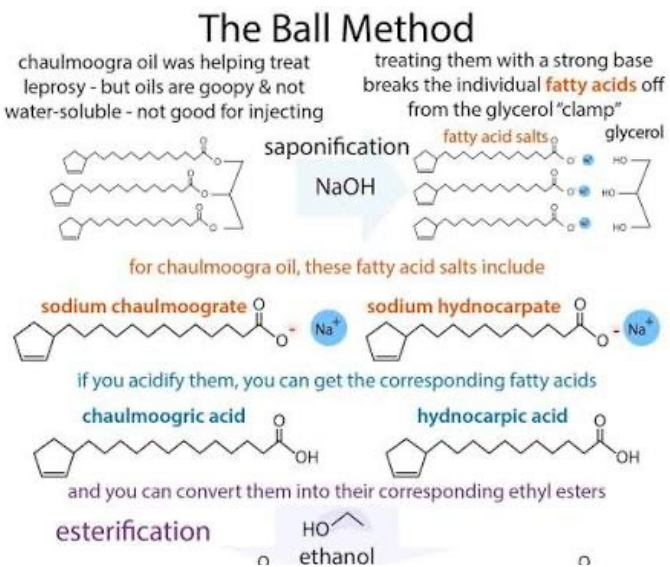
Alice Ball grew up in the 19th century, a time where African Americans faced significant barriers in pursuing education. Despite the discrimination, she earned a double degree in Pharmaceutical Chemistry and Pharmacy from the University of Washington. She excelled in the sciences, achieving impressive feats like being published in the Journal of American Chemical Society for her work on kava plants as an undergraduate.

Ball's achievements earned her a full scholarship to the University of Hawaii for graduate school. She continued studying kava plants, becoming the first African American woman to earn her master's degree from the college. Her research caught the attention of Dr. Harry T. Hollman and Dr. Arthur Dean, both of whom were looking for a cure to Leprosy.

At the time, leprosy was one of the most feared illnesses in the world because it meant lifelong isolation. Patients diagnosed with Leprosy were forced to quarantine away from society because there was no effective treatment. The main medication the doctors offered was chaulmoogra oil. While it seemed to work in theory, the oil was too thick to inject properly and was difficult for the body to absorb.

While most scientists had given up on chaulmoogra oil, Ball spent her time under Dr. Hollman and Dr. Dean researching ways to improve the way the oil was delivered into the body. She found a way to isolate the fatty acids in the oil and chemically modify them into ethyl esters. This created a form of the oil that was injectable and much easier for the body to absorb. This advancement became known as the Ball method and it created the first effective treatment for Leprosy. The death sentence patients with Leprosy were once handed was taken away as millions improved enough to be released from quarantine.

Alice Augusta Ball unfortunately died at only 24 years old. After her death, another scientist took credit for improving the treatment while her contributions were forgotten. Today, however, we're finally able to recognize Ball for her incredible work that cured millions of people with Leprosy.



# Tu Youyou

## The Scientist who Rediscovered a Lost Cure to Malaria



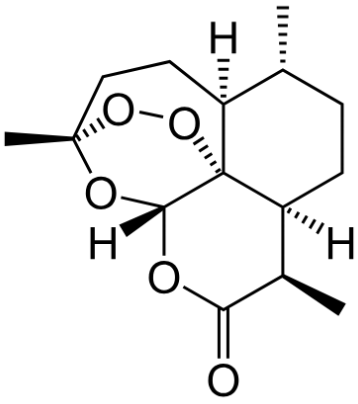
When reading an ancient medical text, we would probably write off its remedies as outdated, useless superstition, or folklore. But every so often, history gets it right. This is the story of Tu Youyou, a scientist who re-discovered a 1700-year-old treatment from ancient China that has since saved millions of lives from malaria.

Tu Youyou was born in 1930 in the coastal city of Ningbo, China. As a child, she was raised around the value of education, working hard throughout elementary and middle school. However, her education was interrupted when she contracted tuberculosis at just 16 years old. Forced to step away from school for 2 years while recovering, she became inspired to study medicine, in the hopes of discovering cures for the same diseases that affected her.

In 1951, Tu Youyou went on to study pharmacology and medicinal plants at Beijing Medical College. After graduation, she continued training at the Academy of Traditional Chinese Medicine, where she learned to combine traditional Chinese medicine with modern principles in biochemistry.

Her skills would later be put to the test when the Chinese government appointed her to Project 523, a top-secret military project. In the midst of the Vietnam War, many Chinese soldiers and their North Vietnamese allies began to contract malaria due to mosquitos, drastically reducing the strength of their armies. In addition, the parasite responsible for malaria, *Plasmodium falciparum*, had grown resistant to typical malaria treatments like chloroquine. As such, the project aimed to discover a new anti-malarial drug.

While others investigated synthetic compounds, Tu Youyou took a different approach, reading ancient medical texts and interviewing other practitioners of traditional Chinese medicine. Eventually, her team learned about the sweet wormwood plant, which was referenced in a 4th-century book to treat malarial fever. After following the preparation process described in the book, the resulting extract successfully treated malaria in experimental monkeys and rats, before curing 21 malarial patients in a 1972 clinical trial. Later work would reveal the active compound in the extract, which was later named qinghaosu, named for the Chinese name for the sweet wormwood plant (qinghao).



Once the project's findings were officially declassified and published to the world, the drug gained its more well-known name, artemisinin, named for the scientific name of the sweet wormwood plant (*Artemisia annua*). The drug quickly became a worldwide sensation, with the United Nations, World Health Organization, and the World Bank each inviting Tu Youyou to share her scientific findings. After rigorous trials and further testing, artemisinin combination therapy became the World Health Organization's official recommendation for malaria treatment. Since then, artemisinin-based combination therapies have been widely used in malaria hotspots such as sub-Saharan Africa and Southeast Asia, and have saved millions of lives globally.

In recognition for what some have touted as "...arguably the most important pharmaceutical intervention in the last half-century", Tu Youyou was awarded the 2015 Nobel Prize in Physiology or Medicine, becoming the first Chinese scientist to win a Nobel Prize in a scientific discipline.

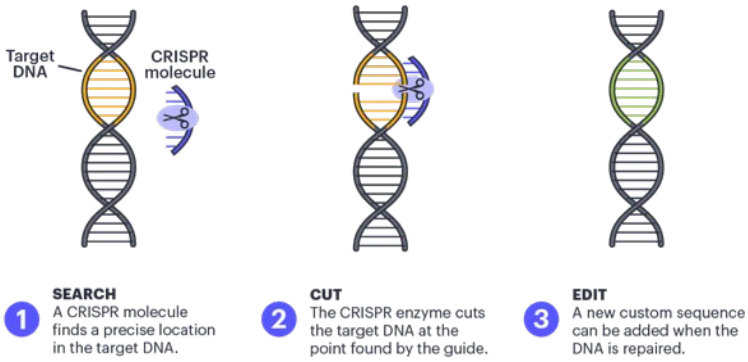
**Discussion:**

- With the rise of Western medicine, alternative therapies are often rejected due to lack of scientific rigor and empirical evidence. However, by rejecting anecdotal evidence, artemisinin would not have been discovered. How can we strike a balance?

# Emmanuelle Charpentier & Jennifer Doudna

## The Unlikely Duo that Discovered Nature's "Copy and Paste"

Imagine taking a walk with a total stranger, sharing one of your imaginations, and winning the most prestigious scientific award with that same stranger! This is the story of Jennifer Doudna and Emmanuelle Charpentier, two female scientists from opposite sides of the world who met each other by total coincidence, joined forces, and proceeded to change the face of modern biology and medicine.




Doudna grew up in the small town of Hilo, Hawaii, while Charpentier grew up just south of Paris. Their shared love for nature and science during childhood led them to pursue further education, respectively earning PhDs in molecular pharmacology and microbiology. Afterwards, Doudna became a leading expert in RNA chemistry, while Charpentier researched bacteria, discovering that their immune system used a special type of RNA called tracrRNA. Later, tracrRNA was found to not only identify foreign genetic material, but also remove it and allow bacteria to repair their own genetic information.

At the time, biologists worldwide were facing a frustrating conundrum. While an organism's genetic information could be read from beginning to end, changing this information was nearly impossible. During experiments, researchers were forced to use methods that were imprecise, slow, and expensive. However, with Charpentier's discovery, scientists realized that nature already had a solution.



By total coincidence, Doudna and Charpentier both flew to Puerto Rico in 2011 to attend the same scientific conference, and met up at a local cafe. As they wandered the streets, they realized they shared an interest in CRISPR-Cas9, a bacterial protein that uses tracrRNA to remove foreign DNA. Charpentier also realized that Doudna's knowledge of RNA chemistry was the missing piece to help understand the chemistry behind this system.

Driven by their shared curiosity, the two women formed a collaboration. Together, they ran many experiments and discovered that besides Charpentier's tracrRNA, another molecule called CRISPR RNA was also needed. Using this knowledge, Doudna and Charpentier created a "hybrid" molecule that contained both tracrRNA and CRISPR RNA. By 2012, their experiments showed that their hybrid RNA molecules were able to cut specific DNA sequences in the correct locations, becoming the first successful demonstration of a DNA editing system under direct human control. In recognition for their discovery, Doudna and Charpentier were awarded the 2020 Nobel Prize in Chemistry, marking the first time a team of two women has won a Nobel Prize.

Without Doudna and Charpentier's invention, the modern research and medical landscape would look very different. Today, the duo's technology is used to cut and modify DNA at specific, precise locations, elevating our understanding of genetics and opening the possibility for genetic conditions to be permanently cured. For countless families and individuals facing devastating genetic diagnoses, hope may be on the way, made possible by Jennifer Doudna's and Emmanuelle Charpentier's exploration of the humble bacteria.

**Discussion:**

- Although CRISPR-Cas9 is being envisioned as a force of good, the technology can also be misused to harm others. What are some ways that CRISPR-Cas9 can be misused, who would be affected, and how should society/researchers try to limit this?
- Scientists from different backgrounds often work together, since they each can share their own perspectives while designing a project. Identify 1 field of science that you are interested in, and think of 5+ fields that would be able to give helpful guidance. (Be creative, these fields can be non-science too!)

**Ingo Potrykus & Peter Beyer: The team that invented a pigmented and vitamin-packed rice.**

Have you ever heard of rice that grows yellow grains and can treat vitamin A deficiency? This is the story of Drs. Ingo Potrykus and Peter Beyer's development of golden rice.



Potrykus and Beyer were both German scientists that graduated from their undergrads with degrees in biology. Potrykus then went on to attain his Ph.D. in biology at the Max-Planck Institute for Plant Breeding Research and Beyer got his Ph.D. in cell biology at the Universities of Marburg and Freiburg. After receiving their degrees, Potrykus went on to serve as a research group leader at the Max-Planck Institute for Genetics & Friedrich Miescher-Institute, then a full professor in plant sciences at the Swiss Federal Institute of Technology while Beyer served as a professor at the Department of Cell Biology at the Centre for Applied Biosciences at the University of Freiburg.



These two studied topics that would mesh perfectly into their final product. Potrykus researched applications of genetic engineering of primarily grain "food security" crops, also known as staple foods. This genetic engineering of plants, which you might now know as genetically-modified organisms, GMOs, was dedicated to combating disease and pests and increasing yield and biosafety. Beyer studied the biosynthesis of carotenoids, naturally occurring orange pigments that act as antioxidants in humans. This means that these compounds protect your cells from oxidative stress and damage that could've led to heart disease or cancers.

When Potrykus and Beyer combined their work, we got the invention of golden rice. It's a type of rice that's enriched with beta-carotene, a type of carotenoid that is converted by the body to vitamin A. The Golden Rice Project, as it's now known, aims to treat vitamin A deficiency in populations around the world. However, the project has faced many hurdles in terms of regulatory approval and prejudice against GMOs. The Golden Rice Project was met with concerns of reducing biodiversity in rice varieties, confusion surrounding its potential health risks, and worries about the socioeconomic implications of implementing golden rice. However,

it was able to receive approval from groups like the FDA, the Bill & Melinda Gates Foundation, Health Canada, and even a group of 107 Nobel laureates, who sent a letter to environmental activist group Greenpeace asking them to accept the genetically modified plant.

While Potrykus and Beyer are still facing obstacles associated with trying to promote widespread adoption of a genetically modified organism, their work is still truly revolutionary in improving the diet and nutrition of people across the world. As for now, I'm looking forward to how synthetic biology and genetic engineering will continue to change the world we know for the better! What do you think?

Hopefully now you can say that you've learned about a scientist you may not have known before!

***Suggested Activity**:* Discussion of natural pigments (like carotenoids, anthocyanins, chlorophyll, betalains, and anthoxanthins) and their effects. Can bring in different fruits and vegetables and discuss what compounds make them the colors they are and how those colors affect their nutrients.

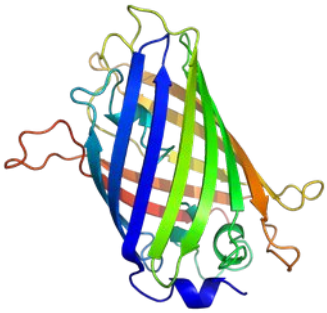
**Osamu Shimomura: The man who discovered the protein that makes genes glow.**

Have you heard of the scientist that survived the bombing of Nagasaki and discovered a glowing protein used in cell and molecular biology? This is the story of Dr. Osamu Shimomura.



Shimomura grew up in Nagasaki, Japan and was only 16 when the bombs dropped. After the fallout, he ended up enrolling at the College of Pharmaceutical Studies at Nagasaki Medical College, which had at the time been uprooted by the blast, to pursue a bachelor's in pharmacy. Upon finishing at Nagasaki, he began working at Nagoya University with Professor Yoshimasa Hirata, eventually receiving his master's and Ph.D. in organic chemistry, studying what compound made sea-fireflies glow when in contact with water.

His research interested Professor Frank Johnson at Princeton University in New Jersey, where he was then recruited to. While at Princeton, he studied crystal jelly, a type of bioluminescent jellyfish, working at the University of Washington's Friday Harbor Laboratories in the San Juan Islands of Washington state. There, he discovered a protein called green fluorescent protein, or GFP, that glows green when exposed to blue or UV light. In his work for the discovery and development of GFP, he, along with colleagues Roger Y. Tsien and Martin Chalfie, were awarded with the Nobel Prize in Chemistry in 2008.



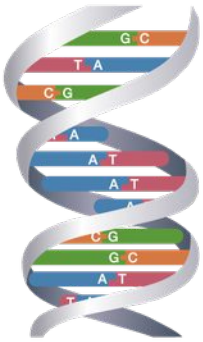
Shimomura's discovery is still used today, over 60 years later, in order to help scientists study protein expression. By attaching the gene for GFP to genes of interest in DNA, whenever that gene is turned on, so is GFP, giving off a bright green glow. Because GFP has no other effect on its host when expressed and can be visualized noninvasively, meaning scientists can see it without physically cutting into or entering the organism, it's perfect for studying genes of interest without adverse effects on the host.

Shimomura's efforts were a breakthrough in the world of cell and molecular biology and serves as a story of the indomitable human spirit. Hopefully now you can say that you've learned about a scientist you may not have known before!

***Suggested Activity*:**
Educators can teach students about transforming *E. coli* with GFP. Many companies have kits for students to work with like the Bio-Rad pGLO Bacterial Transformation Kit that are available for purchase. This activity ties into discussions of plasmids and bacterial transformation, gene expression, proteins.

# Daisy Roulland-Dussoix

## Discovery of Restriction Enzymes



DNA is like an instruction manual for living things. It contains the information cells use to build proteins, which allow them to grow and carry out different functions. In synthetic biology, scientists often want to change these instructions. This could look like adding a gene to allow bacteria to make a useful medicine, remove a gene that causes a problem, or combine DNA from different sources to give a cell a new function. But DNA is extremely long, how do scientists cut where they want?



In the 1960s, Daisy Roulland-Dussoix was a young scientist working with Werner Arber to understand how bacteria defended themselves against viruses. They were puzzled why some bacteria could resist viral infection. To figure out what was happening, Daisy performed experiments that allowed her to follow the virus's DNA after it entered the bacterial cell. Expecting to learn why the virus couldn't reproduce, Daisy was shocked to find the viral DNA was being cut into pieces.

Daisy found that bacteria have a natural defense system against viruses which recognizes foreign DNA and uses special proteins to cut it apart. These proteins became known as **restriction enzymes**, tiny molecular scissors that recognize particular DNA sequences and cut at those locations.

Scientists eventually realized that they could borrow these molecular scissors from bacteria, and retarget them to different genes in the laboratory. Restriction enzymes gave researchers one of the first reliable ways to cut DNA exactly where they wanted, allowing scientists to combine DNA, move genes between organisms, clone DNA, and even create medicine like recombinant insulin.

What makes Daisy's story especially interesting is that her discovery is not what she initially expected. She began by asking why a virus behaved differently in different bacteria, but passionately following that question helped uncover a tool that would transform synthetic biology and genetic engineering. Science is weird that way. The most important discovery could be something you weren't even looking for.

**Discussion Questions:**

- Daisy Roulland-Dussoix discovered something important while investigating a different question. What does her story teach us about the role of curiosity and unexpected results in science?
- Restriction enzymes began as a defense system in bacteria but were transformed into tools using synthetic biology. What are some other examples of things in nature that humans have adapted into useful technologies?

# Daniel Hale Williams

## The Pioneer of Open Heart Surgery



Today, if someone were to suffer a serious injury to the heart, surgeons can use medical technologies such as imaging, antibiotics, and anesthesia to ensure the heart is well characterized and the internal environment is suitable for operation. In the late 1800s, none of these existed. Doctors couldn't see the chest, infection was a major cause of death, and surgery to the heart was so dangerous that most textbooks simply stated that it should not be attempted.




In 1893, this was the situation Daniel Hale Williams found himself in after a man named James Cornish was brought to Chicago's Provident Hospital after being stabbed in the chest during a fight. Mr. Cornish was losing blood quickly and struggling to breath, leading Dr. Williams suspected that the knife had damaged his heart. No imaging technology could tell Dr. Williams knew exactly what had happened, the only way to find and repair the injury to save Mr. Cornish's life was to operate.

Dr. Williams opened Cornish's chest, removed part of a rib so he could directly reach the heart, and discovered a damaged blood vessel and a wound in the **pericardium**, the protective sac around the heart. Working while Cornish's heart was still beating, Dr. Williams carefully **sutured**, stitched closed, the damaged tissue.

Cornish survived and after a mere 51 days, was able to leave the hospital, and went on living more than 20 years. William's operation became the earliest successful surgery involving the heart and helped challenge the belief that injuries near the heart were essentially untreatable.

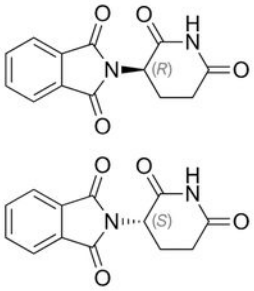
Dr. Williams would go on to serve as surgeon-in-chief at Freedmen's Hospital in Washington, D.C., where he worked to improve surgical standards and medical training. He also helped co-found the National Medical Association, creating a professional organization for Black physicians who were often excluded from existing medical societies. Combined with his work building hospitals, training providers, and challenging racial barriers in medicine, Dr. Williams helped expand both what surgeons believed was medically possible and who could participate in medicine at all.

**Discussion Questions:**

- Daniel Hale Williams performed this surgery without many of the technologies surgeons rely on today. What qualities or ways of thinking do you think allowed him to attempt such a dangerous feat?
- Dr. Williams' impact came both from advancing surgery and from creating opportunities for Black doctors and nurses. How can scientific innovation not only improve medicine, but also who can participate and benefit from it?

# Frances Oldham Kelsey

## The Doctor who saved America from Thalidomide




When talking about chirality it's important to discuss its importance in biological systems. The most glaring example of this, is the story of Thalidomide. Thalidomide was a drug sold in the UK and Canada in the 1950s and 60s to treat morning sickness. What was advertised as one drug was actually 2 stereoisomers of the same drug. While one did help with morning sickness, the other caused children to be born with missing or deformed limbs. This led to many deaths in the coming decades, and seemingly no one knew what caused this.



In America, the new Doctor Frances Oldham Kelsey was appointed as one of only 7 doctors reviewing drug applications in the FDA full time. She had previously gotten her PhD. working on synthetic drugs to treat malaria, discovering that many pass through the placental barrier that separates fetus from mother.

Kelsey quickly found her studies helpful when the application for Thalidomide came across her desk. She quickly realized that the drug was chiral, but didn't see any mention of it in the papers. She quickly dug deeper and found a number of flawed studies, and a history of unscrupulous data.

At the time, the FDA was not allowed to outright reject drug applications, but only request more information every 60 days. So she sent back the application every 60 days for over a year. Drawing on her training as a chemist and a scientist, she recognized that thalidomide passed the placenta barrier, and it was eventually revealed that unpublished mouse studies found birth defects. In this time, the company spread rumours that Kelsey was pushing back on the drug out of sheer laziness rather than an abundance of caution.

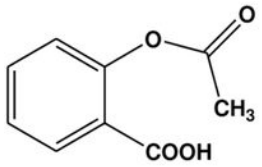
If it were not for the work of Frances Oldham Kelsey, the America we know today would look a lot different. It is because of her that the FDA now has the power to reject therapeutics from coming to the market. Many parents, loved ones, and friends may not have been alive if Thalidomide came to the U.S. Many still deal with the consequences of Thalidomide today, simply because their parents took a drug that many thought was harmless.

**Recommended Activity**
Break up into smaller groups and discuss:

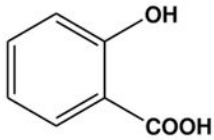
- Why is learning about chirality important for many people beyond just chemists?
- Who does thalidomide impact, and why does that shape our perspective on it?

# Arthur Eichengrün

## The Unknown Scientist Behind Aspirin




Aspirin (prodrug)




Salicylic acid

What we call "Aspirin" is a short-hand name for a molecule called acetylsalicylic acid. Aspirin was originally invented as an alternative to salicylic acid–a molecule commonly found in willow tree root bark. It was known throughout history for its pain relieving effects but as modern medicine progressed, doctors started to notice a number of side effects such as gastric irritation, nausea, or tinnitus. Thus, many chemical manufacturers in the 1890s had interest in developing an alternative.

Introducing Friedrich Bayer & Co, a massive German dye company. Many dye companies had heaps of chemists, and the institutional backing to produce a wide variety of chemicals, so many of them quickly shifted to producing novel medicines. Bayer employed many chemists, namely Felix Hoffmann and Arthur Eichengrün. Eichengrün led the charge towards developing a salicylic acid alternative, and under him, Hoddmann conducted many of the reactions.

For most of history Felix Hoffmann was believed to have first synthesized aspirin in 1897. And while Hoffmann did conduct lab work that yielded pure aspirin, he did so under the direction of Arthur Eichengrün. Eichengrün claimed on several occasions that Hoffmann didn't know the broader purpose of the experiments he was running, and that he should have been given credit for this invention. Although many historians would pass this off as a credit in leadership, Eichengrun wasn't acknowledged for that either.

Eichengrün was a Jewish man, and after Hitler came to power in Germany, he was robbed of many official positions in academia. He was arrested twice and sent to the concentration camp Theresienstadt for 14 months until the end of WWII. Because of the regime at the time, many believe now that his lack of acknowledgements weren't a product of Hoffman's work, but of the Nazi's attempt to remove Jewish scientists from the spotlight. Although we have a better understanding of Eichengrun's contributions to Aspirin today, it begs the question of how many scientists fall through the cracks of documented history.



**Discussion Questions:**

- Who do we typically give credit for scientific advancements?
- How can we acknowledge power imbalances in leadership and historical inequities in cases such as this?